GCSE · OCR · Computer Science

2 hours 39 questions

Networks & Topologies

Networks & Topologies

Types of Network / Factors of Network Performance / Client Server & Peer to Peer Networks / Local Area Network Hardware / The Internet / Star & Mesh Networks

Easy (13 questions)	/20
Medium (12 questions)	/44
Hard (14 questions)	/46
Total Marks	**/110**

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Easy Questions

1 A law company currently use a **Local Area Network** (LAN) **linked** to a **Wide Area Network** (WAN).

Define what is meant by a **Wide Area Network.**

..... **(1 mark)**

2 (a) A university has buildings in two sites that are 5 miles apart.

Describe the **difference** between a **LAN** and a **WAN**.

.....

..... **(2 marks)**

(b) Site A has 4 classrooms. Site B has 2 classrooms. The network on each site between the classrooms is a **star topology** using a **switch**. The two sites are connected over the Internet.

Complete the network diagram for **site A** of the University.

(2 marks)

3 Devices on a network do not currently have Internet access.

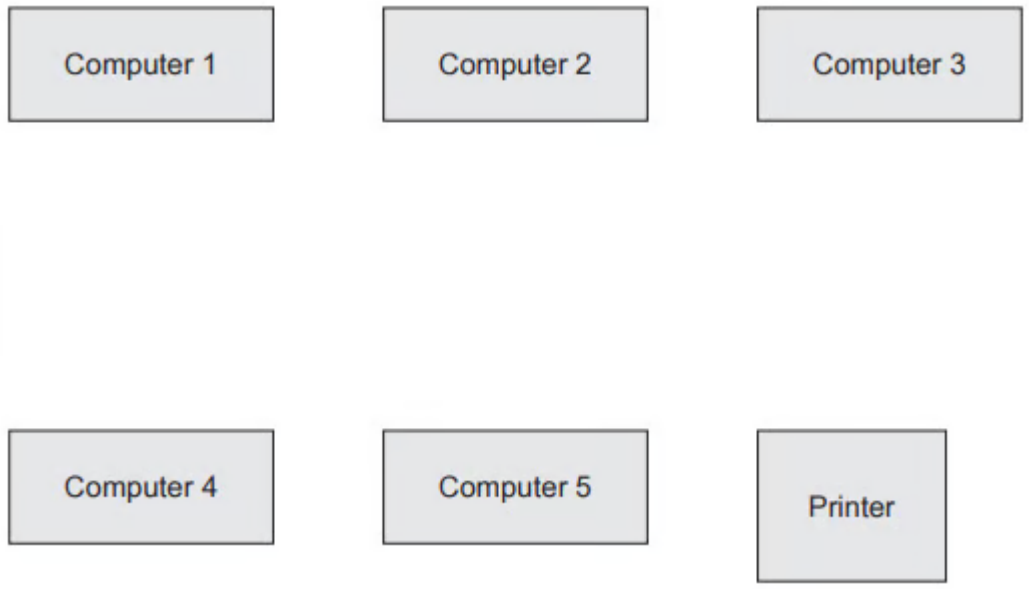
Identify one device that can be used to connect a **home network** to the **Internet**.

..... **(1 mark)**

4 Naomi's office has **five computers** connected into a Local Area Network (LAN). There is also **one printer** that all the devices can print to.

The LAN is set up as a **mesh topology**

Draw connections to show one way that the **devices could be connected** using a **mesh topology**.



(2 marks)

5 An office has a LAN (Local Area Network). The office has **four employees** who **each have a laptop**.

The office also has **one server** and **one networked printer**.

The office is set up as a **star network** with **a switch at the centre**.

All devices are connected to the network using cables.

Draw the **devices and connections** in the office **star network**.

All devices must be clearly labelled.

(3 marks)

6 A user enters a uniform resource locator (**URL**) into a **web browser** on one of the computers in the house.

A system is then used **to find the IP address** of the **web server** associated with the URL.

Name the system which matches **URLs** to **IP addresses** on the web.

..... **(1 mark)**

7 **State** the typical geographical range of a Local Area Network (LAN)

..... **(1 mark)**

8 **Identify** two factors that can affect the performance of a network.

.....

..... **(2 marks)**

9 **Define** the term 'Bandwidth'

..... **(1 mark)**

10 **Identify** the role of a 'client' in a client-server network

..... **(1 mark)**

11 **State** one characteristic of a Peer-to-Peer (P2P) network

..... **(1 mark)**

12 **Define** the term 'Cloud storage'

..... **(1 mark)**

13 **Identify** which protocol is used by a web browser to request a page from a web server

..... **(1 mark)**

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Medium Questions

1 (a) A library has a **LAN** (Local Area Network).

The LAN allows access by both wired and wireless devices. Users have reported that the **network sometimes runs very slowly**.

Explain why the **number of devices using the network at the same time** can **affect the performance** of the network.

.....

.....

..... **(3 marks)**

(b) **Identify one other factor** that can **affect the performance** of the network

..... **(1 mark)**

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2 (a) **Explain two advantages** to a law company of **storing their data** in the **Cloud**.

.....

.....

.....

..... **(4 marks)**

(b) **Explain two disadvantages** to a law company of **storing their data** in the **Cloud**.

.....

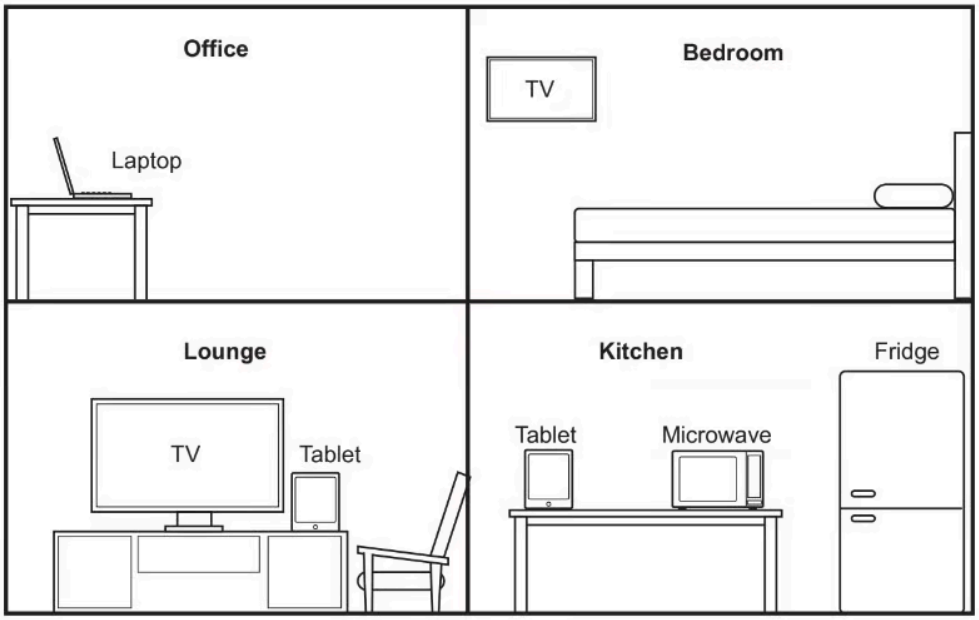
.....

.....

..... **(4 marks)**

3 Hope has a network in her house. The main devices are shown in the diagram.

State whether Hope's network is a **LAN or a WAN**.

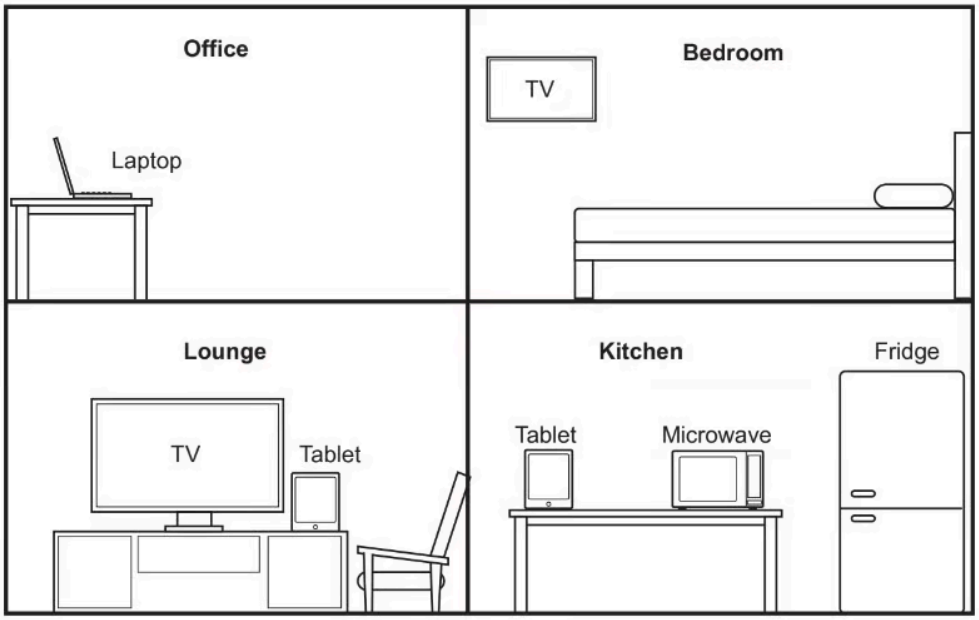


Justify your choice.

.....

..... **(2 marks)**

4 Hope has a network in her house. The main devices are shown in the diagram.



Explain why Hope's network uses a **peer-to-peer model** and not a **client-server model**.

.....

.....

..... **(3 marks)**

5 **Describe** the **role of the switch** in an office network.

.....

..... **(2 marks)**

6 (a) The office introduces a WAP (**Wireless Access Point**) to allow network access to wireless devices.

The office manager has noticed that the **performance of the network has recently decreased**.

Describe how **introducing wireless** access could have **slowed down the network**.

.....

..... **(2 marks)**

(b) **Identify two** other factors that can **affect the performance** of a network.

.....

..... **(2 marks)**

7 A **house** has computers in each room and a central router.

Every room allows both **Ethernet** and **WiFi** connections to the **router**.

Identify if the house network is a **LAN** (local area network) or a **WAN** (wide area network).

Justify your choice.

.....

.....

..... **(3 marks)**

8 (a) **Describe** the **purpose of the router** in a home network.

.....

..... **(2 marks)**

(b) **Identify two additional** items of **network hardware**, apart from cables and a router, that may be used within the house network.

.....

..... **(2 marks)**

9 (a) One office in the airport has five computers connected to one switch. There are two printers in the office that can be accessed by all computers.

The computers are connected using a star topology.

Draw a diagram to show how the five computers, switch and two printers are connected in a star topology

(3 marks)

(b) **Give** one benefit and one drawback of the office using a star topology instead of a mesh topology.

.....

..... **(2 marks)**

(c) **Describe** the role of the switch in the star topology.

.....

.....

..... **(3 marks)**

10 **Describe** two benefits of using a client-server model instead of P2P

.....

..... **(2 marks)**

11 **Explain** the purpose of a Domain Name Server (DNS)

.....

..... **(2 marks)**

12 **Describe** the request-response cycle between a web client and a web server

.....

..... **(2 marks)**

Hard Questions

1 A network has several routers.

Identify three tasks carried out by a **router**.

.....

.....

..... **(3 marks)**

2 Give **two reasons** why a bakery may use a **star network** topology for their LAN.

Reason 1	
Reason 2	

.....

..... **(2 marks)**

3 Site B has a **higher network performance** than site A.

Explain how each of the following can **contribute to the performance** of a network.

- Wifi frequency
- Interference
- Number of concurrent users
- Type of network traffic

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..... **(4 marks)**

4 Hope's files are stored on the cloud.

Describe the benefits and drawbacks to Hope of **storing her files** on the **cloud**.

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..... **(6 marks)**

5 Computer 1 enters the **URL** www.ocr.org.uk into a web browser. This is then converted into the **IP address** of the webserver that hosts the website.

Explain how the **URL** www.ocr.org.uk is **converted** into the I**P address**

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.....

..... **(3 marks)**

6 The IP address 192.149.119.226 is linked to the website with a URL of https://www.ocr.org.uk

When https://www.ocr.org.uk is entered into a browser, the website homepage is loaded.

Describe the **relationship between** the website **URL** (https://www.ocr.org.uk), the **IP address** and the **webserver**.

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..... **(5 marks)**

7 A university has multiple campuses across a city.

Explain why the connection between these campuses is a WAN, referring to **ownership** of the infrastructure.

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..... **(3 marks)**

8 A bank uses a WAN to connect its branches.

Discuss the security implications of using a WAN compared to a LAN.

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..... **(3 marks)**

9 **Explain** how an increase in the number of users can lead to a decrease in network performance

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.....

..... **(3 marks)**

10 A small creative agency uses a P2P network.

Evaluate this choice as the agency expands to 50 employees and multiple rooms.

.....

..... **(2 marks)**

11 **Compare** the role of a switch and a router within an airport network.

.....

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..... **(4 marks)**

12 **"A DNS lookup for a website fails."**

Explain how a user can still access the website if they know its IP address

.....

.....

..... **(3 marks)**

13 A company uses Cloud services for its office software.

Analyse the impact if their Internet connection fails.

.....

..... **(2 marks)**

14 **"A user types a website URL into their browser's address bar."**

Explain how the browser determines the location of the web server to connect to it and display its web page to the user

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.....

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(3 marks)